MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, INTERFACE, DIGITAL, KEY CODE EKB

This amendment forms a part of Military Specification MIL-M-28787/162(EC), dated 17 March 1978, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Bottom of page: Delete "FSC MISC" and substitute "FSC 5963".

1.2, Environment: Delete "Class I as specified in MIL-M-28787" and substitute "Class I as specified in MIL-STD-1389".

PAGE 3

1.3: Delete the first sentence in its entirety.

PAGE 5

3.5.8, second line: Delete "M28787/162" and substitute "M28787/162-1".

PAGE 9

TABLE II, Test conditions R1-R8, S1-S8, and T, Test type column: Delete "X" (three places) and substitute "S" (three places).

* TABLE II, Test condition T, Parameter column: Delete ", C_L = 50 pF".

* TABLE II, Test condition T, Ref test table column: Delete "XVIII" and substitute "XVII".

PAGE 11

* 4.3: Delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGE 13

Add new paragraph 6.5 as follows:

"6.5 Cross reference of classification. Part number M28787/162-1 supersedes part number M28787/162. The two parts are interchangeable."

PAGE 22

* FIGURE 7: Delete and substitute new figure 7 as printed on page 3 of this amendment.

PAGE 27

* FIGURE 9: Delete and substitute new figure 9 as printed on page 4 of this amendment.

PAGE 28

* FIGURE 10: Delete and substitute new figure 10 as printed on page 5 of this amendment.

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PAGE 30

* FIGURE 11: Delete and substitute new figure 11 as printed on page 6 of this amendment.

PAGE 41

* FIGURE 16: Delete and substitute new figure 16 as printed on page 7 of this amendment.

PAGE 43

* FIGURE 17: Delete and substitute new figure 17 as printed on page 8 of this amendment.

PAGE 50

* TABLE XVII, Parameter column: Delete ", C_L = 50 pF".

PAGE 51

* FIGURE 21: Delete and substitute new figure 21 as printed on page 9 of this amendment.

PAGE 52

* FIGURE 22: Delete and substitute new figure 22 as printed on page 10 of this amendment.

PAGE 54

* FIGURE 23: Delete and substitute new figure 23 as printed on page 11 of this amendment.

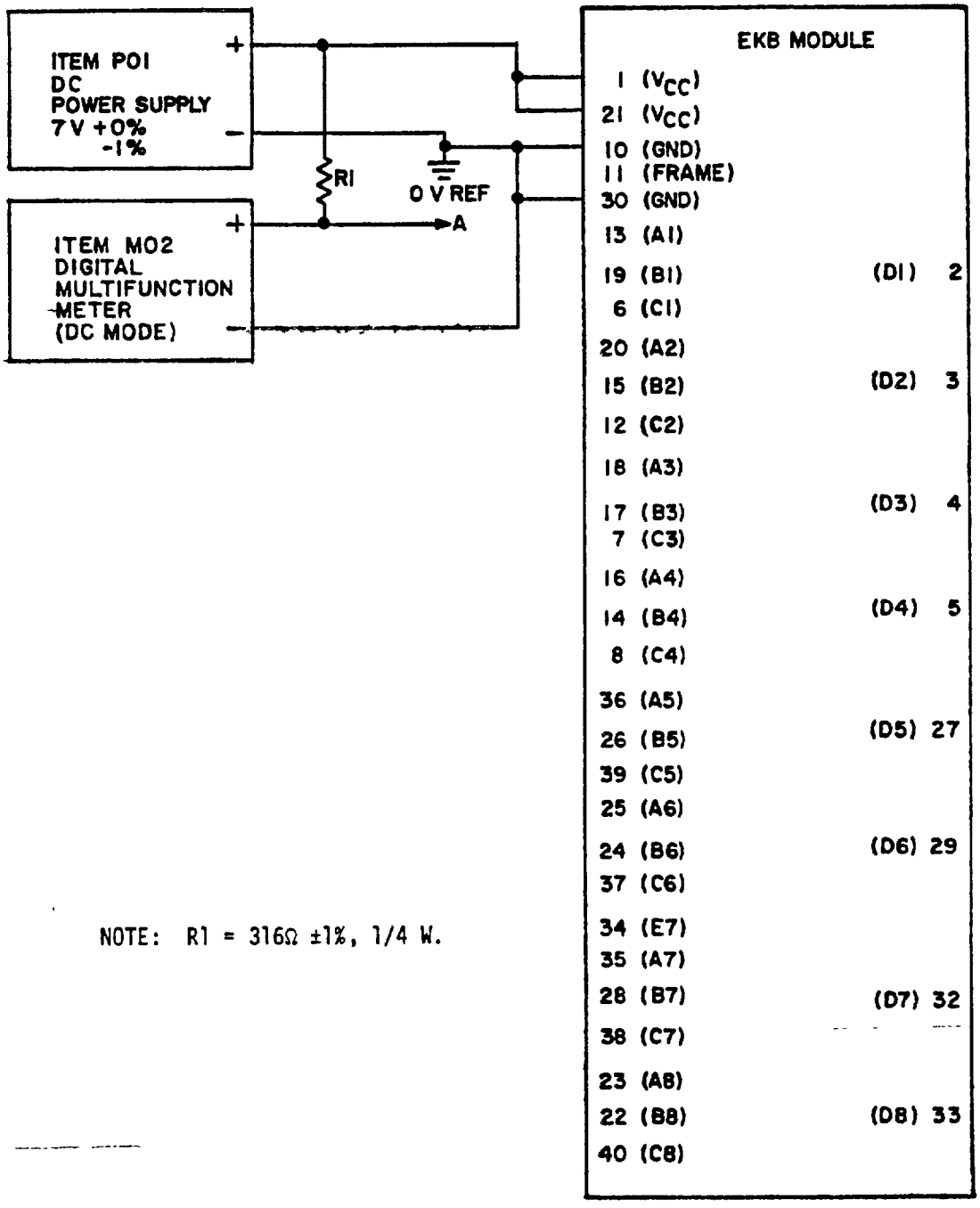
NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Agent:
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Navy - EC

(Project 5963-N592)

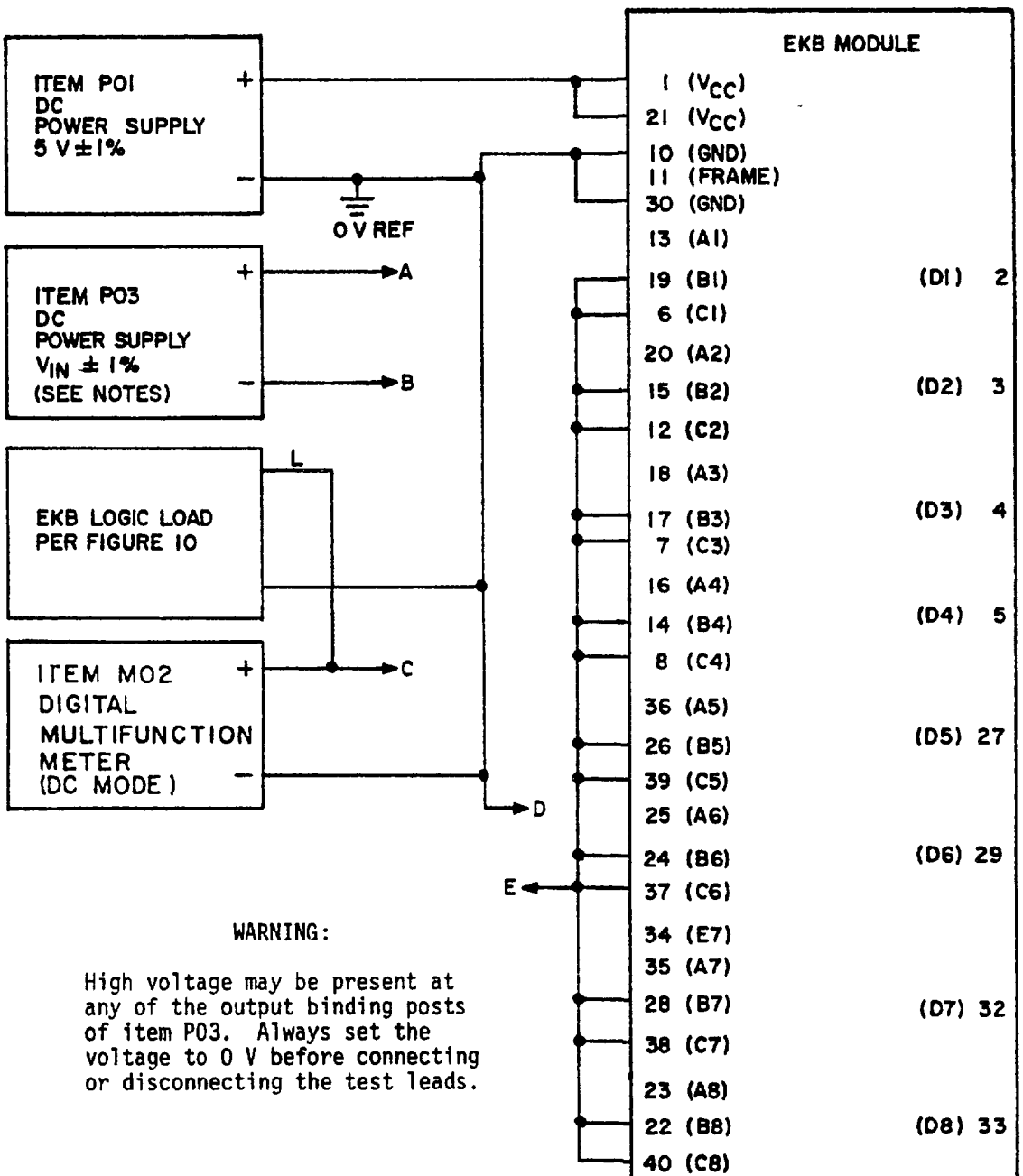
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NOTE: R1 = 316Ω ±1%, 1/4 W.

FIGURE 7. V_{out} (low loaded) test circuit for table IV.

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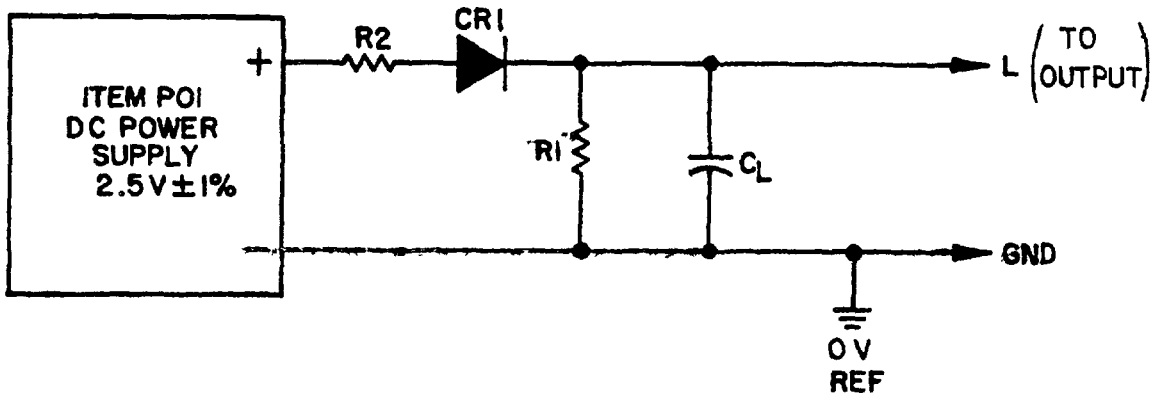
WARNING:

High voltage may be present at any of the output binding posts of item P03. Always set the voltage to 0 V before connecting or disconnecting the test leads.

NOTES:
1. For test conditions D1-D16, V_{IN} = 100 V ±1%.
2. For test conditions D17-D32, V_{IN} = 300 V ±1%.

FIGURE 9. Common mode voltage test circuit for table VI.

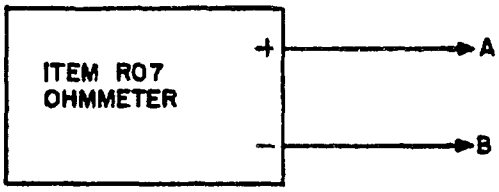
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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 3.01 kΩ ±1%, 1/4 W.
3. R2 = Select value for a current flow out of point L of 16 mA ±1% with point L held at 0.4 V ±1%.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P1-P8, Q1-Q8, R1-R8, S1-S8, U1-U8, V1-V8, and W1-W8.
5. C_L is not required for test conditions D1-D32, L1-L8, and M1-M8.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 10. EKB logic load circuit.

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NOTE: For test condition E10, penetrate the anodized finish on the module fin with a sharply pointed probe connected to the negative terminal.

EKB MODULE		
1 (V_{CC})		
21 (V_{CC})		
10 (GND)		
11 (FRAME)		
30 (GND)		
13 (A1)		
19 (B1)	(D1)	2
6 (C1)		
20 (A2)		
15 (B2)	(D2)	3
12 (C2)		
18 (A3)		
17 (B3)	(D3)	4
7 (C3)		
16 (A4)		
14 (B4)	(D4)	5
8 (C4)		
36 (A5)		
26 (B5)	(D5)	27
39 (C5)		
25 (A6)		
24 (B6)	(D6)	29
37 (C6)		
34 (E7)		
35 (A7)		
28 (B7)	(D7)	32
38 (C7)		
23 (A8)		
22 (B8)	(D8)	33
40 (C8)		

FIGURE 11. Resistance and continuity test circuit for table VII.

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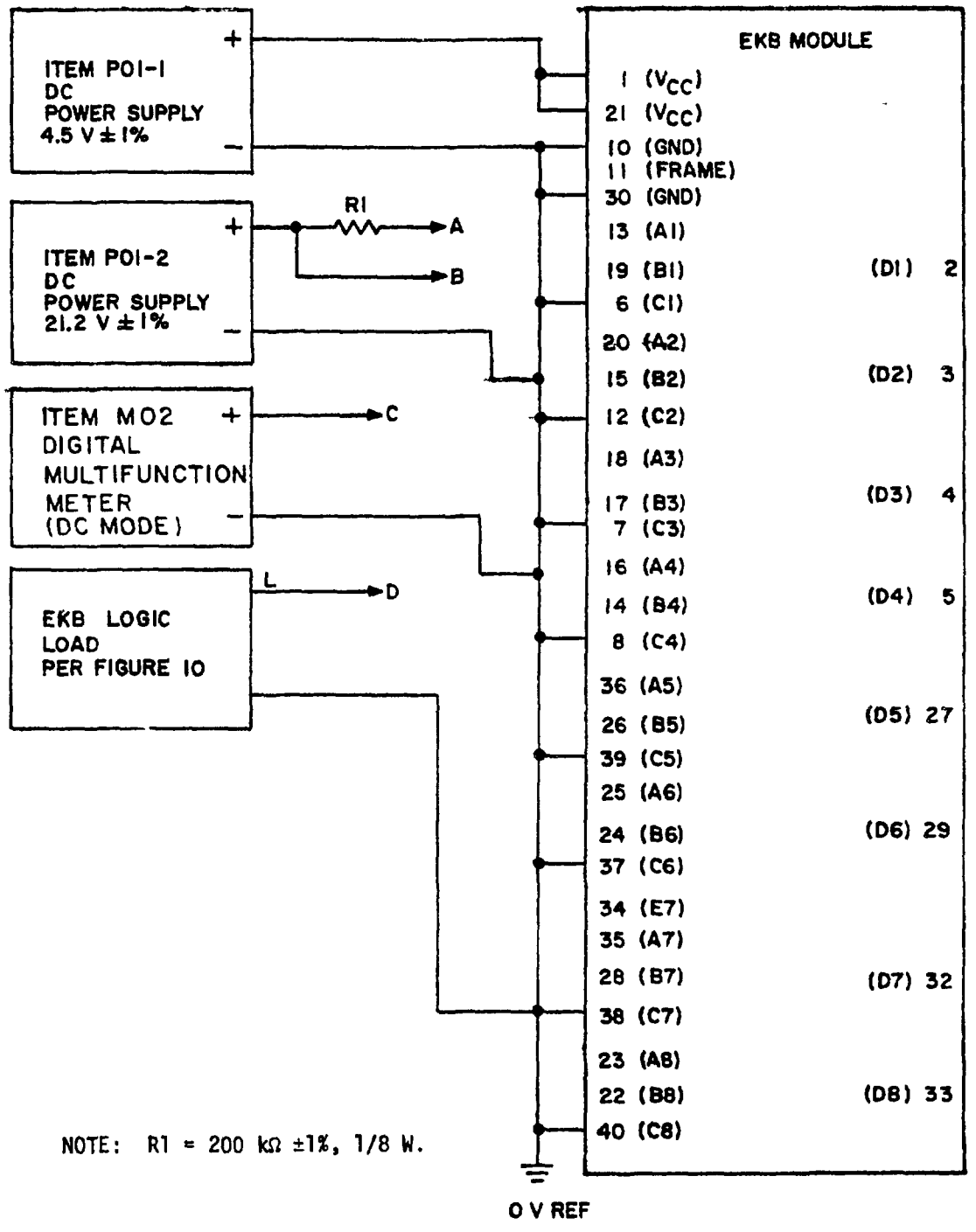


FIGURE 16. Input voltage, low level, B/C input (I_{IL} = 100 µA) and output voltage, low level, I_{IL} = 100 µA, test circuit for tables XII and XIII.

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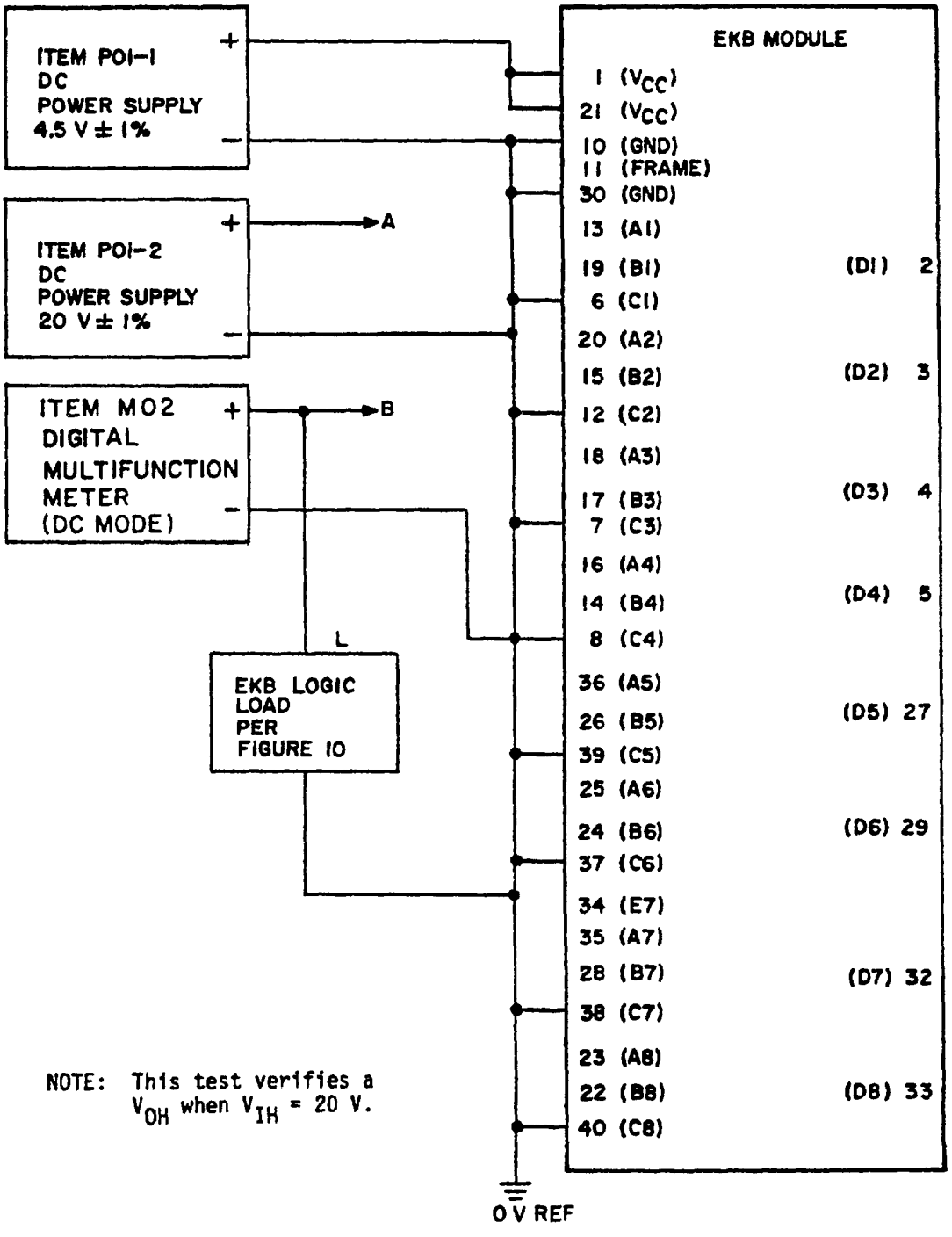
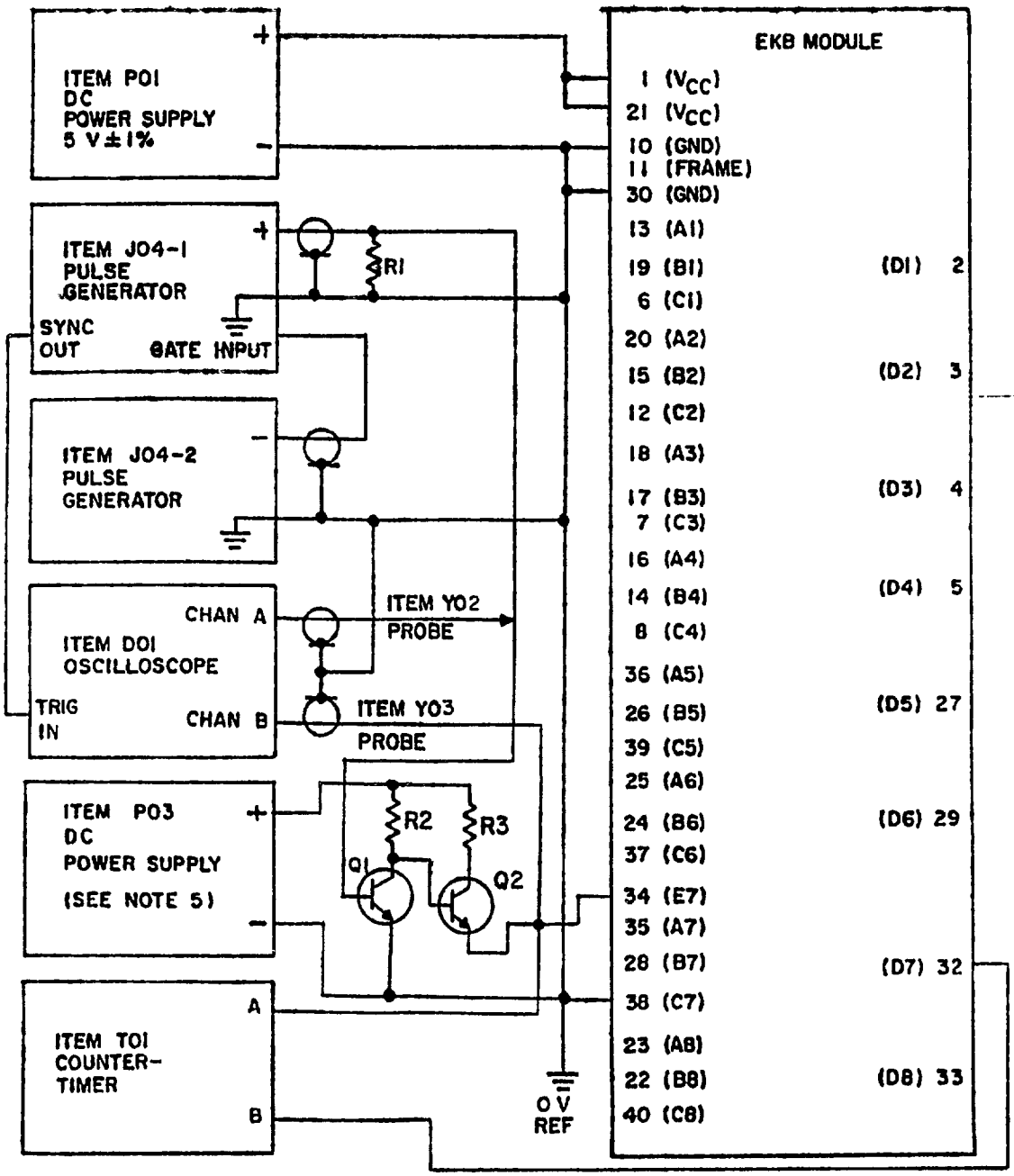


FIGURE 17. Output voltage, high level, V_{IH} = 20 V, test circuit for table XIV.

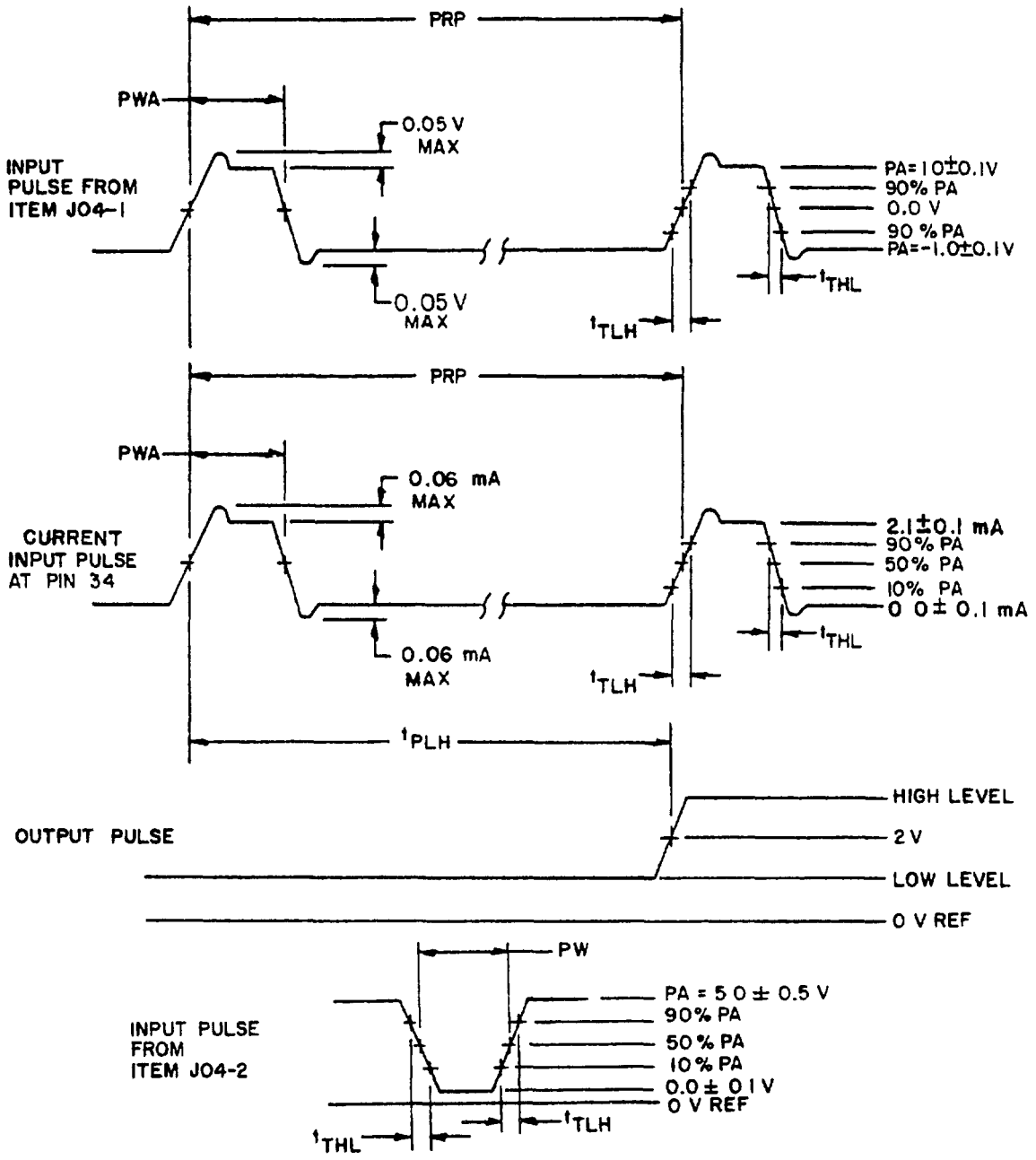
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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. R2 = 40 kΩ ±1%, 2 W.
3. R3 = 49.9 kΩ ±1%, 1/2 W.
4. Q1 = Q2 = 2N3738, or equivalent.
5. Adjust item J04-1 and P03 before connecting item J04-2 output to item J04-1 gate input. Set item P03 initially at 200 V dc and adjust after item J04-1 has been connected to obtain input current pulse at pin 34.
6. After making circuit connections, apply one pulse with item J04-2 and make measurement.

FIGURE 21. Propagation delay time (t_{PLH}), E/C input, 400 Hz, test circuit for table XVII

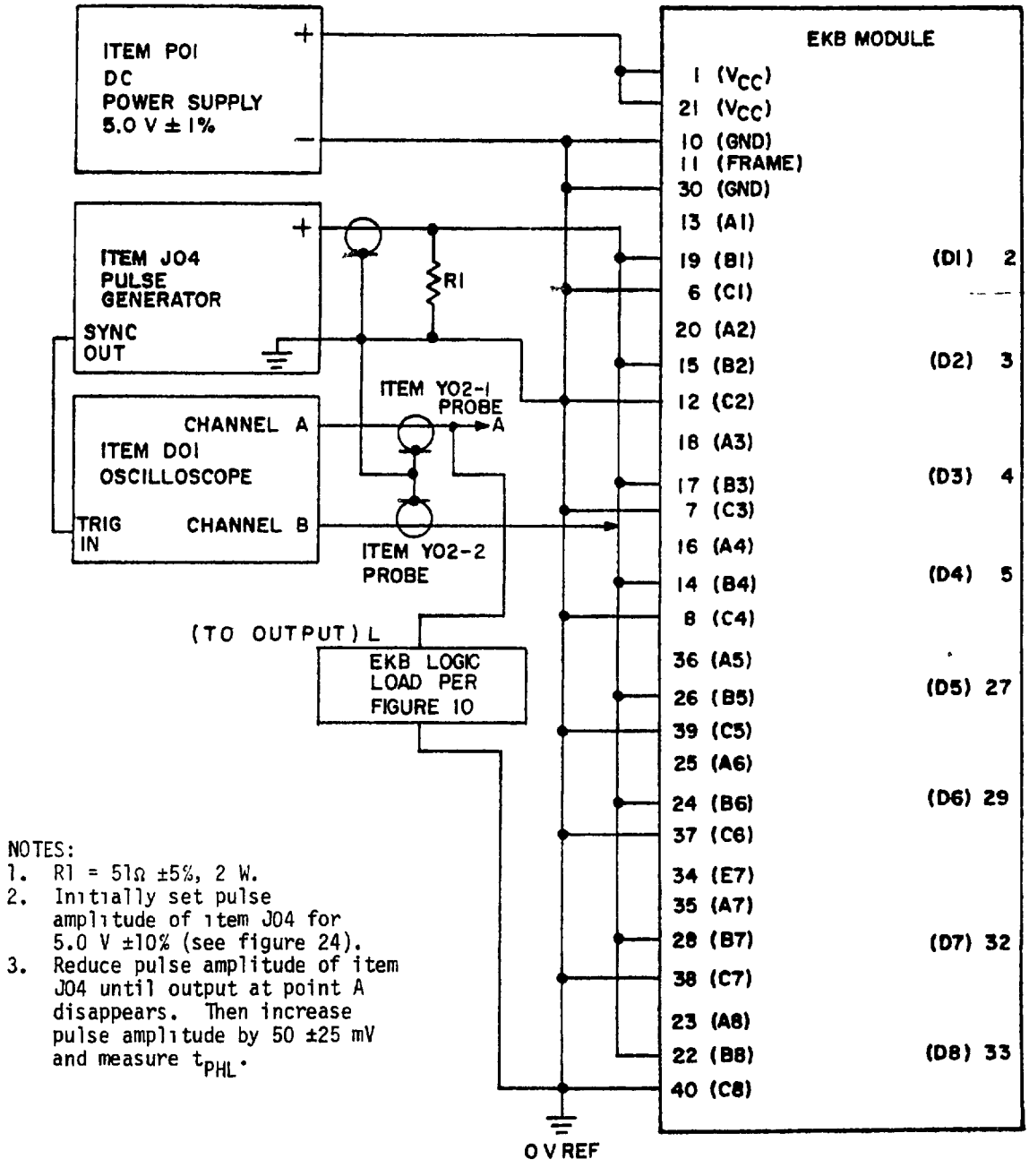
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NOTES:
1. Input pulse parameters for item J04-1:
   PWA = 0.95 ±0.05 ms.
   PRP = 2.5 ±0.1 ms.
   $t_{TLH} = t_{THL} = 5 \pm 2$ µs.
2. Input pulse parameters at pin 34:
   PWA = 0.95 ±0.05 ms.
   PRP = 2.5 ±0.1 ms.
   $t_{TLH} = t_{THL} < 10$ µs.
3. Input pulse parameters for item J04-2:
   PRP = 240 ±40 ms.
   PW = 120 ±20 ms.
   $t_{TLH} = t_{THL} = 100 \pm 10$ µs.
4. After setting up the pulse, place item J04-2 in the single pulse mode.

FIGURE 22. Propagation delay time (t_{PLH}), E/C input, 400 Hz, test circuit waveforms for table XVII.

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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. Initially set pulse amplitude of item J04 for 5.0 V ±10% (see figure 24).
3. Reduce pulse amplitude of item J04 until output at point A disappears. Then increase pulse amplitude by 50 ±25 mV and measure t_{PHL}.

FIGURE 23. Schmitt trigger hysteresis (measuring propagation delay time t_{PHL}) check test circuit for table XVIII.

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